BrailleAI

Bidirectional Braille AI Communicator

I. Core Operating Modes

The system acts as a unified platform delivering three distinct AI-driven functionalities:

- **Mode 1: Face-to-Face Conversation:** A bidirectional communication bridge between a DeafBlind user and a hearing person. The user's Braille input is converted to text-to-speech (TTS), while the hearing person's spoken responses are captured via speech-to-text (STT) and processed by an LLM. The AI summarizes the response and translates detected emotional tone into specific tactile Braille prefix symbols.
- **Mode 2: AI Braille Tutor:** An adaptive educational interface for learning Braille. The system generates curriculum challenges, verifies the user's keyboard inputs, provides immediate tactile and audio feedback, and tracks learner progression data like accuracy and confusion pairs.
- **Mode 3: AI Assistant:** A standalone, silent LLM interface. It processes tactile queries and delivers answers directly to the Braille display, utilizing predictive response suggestions to increase conversational bandwidth.

II. Hardware Architecture

The physical build utilizes a servo-based mechanical design to minimize cost, wiring complexity, and power draw.

- **Microcontroller:** ESP32-S3 WROOM module, responsible for handling I2S audio, PWM signals, SPI communication, and Wi-Fi requests.
- **Braille Output (Actuators):** Two 6-dot Braille cells driven by four MG90S micro servos.
- **Braille Output (Mechanics):** Each servo controls a 3D-printed linear slider featuring 8 distinct cross-sectional bump profiles. As the servo rotates, the slider shifts linearly to mechanically push specific guide pins up or let them drop down, holding the state without active current.
- **Braille Input:** A 10-key Perkins-style tactile keyboard (6 dot keys, space, enter, and mode keys) wired directly to GPIO pins, utilizing the ESP32's internal pull-up resistors for chord detection.
- **Audio Peripherals:** An INMP441 MEMS microphone for I2S audio capture, and a MAX98357A DAC/amplifier paired with a speaker for I2S audio playback.
- **Data Logging:** A micro SD card module connected via the hardware SPI bus.

III. Firmware & Software Stack

The on-device logic manages peripheral communication, user inputs, and hardware coordination.

- **Core Framework:** Written in C++ using FreeRTOS for asynchronous task management and state machine routing between the three modes.
- **Input Processing:** Debounced chord detection logic that registers simultaneous key presses within an 80ms window and decodes them into ASCII characters using a Grade 1 Braille lookup table.
- **Output Processing:** Translates standard ASCII characters into 6-bit Braille patterns, maps those to left and right half-cell states (0-7), and translates the states into specific PWM angles for the servos.
- **Local Logging:** Writes structured, newline-delimited JSON entries to the SD card to track conversational transcripts, tutor mastery data, and API latency metrics.

IV. Cloud & AI Infrastructure

The backend relies entirely on external APIs handled directly by the ESP32's HTTP client.

- **Speech-to-Text (STT):** Wit.ai or OpenAI Whisper API for converting captured I2S microphone audio into text.
- **Text-to-Speech (TTS):** Google Cloud TTS to synthesize spoken audio from the user's typed Braille.
- **Language & Intelligence (LLM):** Anthropic Claude Sonnet 4.6 handles all complex reasoning. This includes parsing conversational tone for emotion prefixes, executing context-aware summarization, and running the adaptive tutor logic.
- **Cloud Synchronization:** When Wi-Fi is available, the ESP32 pushes the SD card's JSON logs to a Google Apps Script webhook, which automatically appends the data to a Google Sheet for remote monitoring by teachers or therapists.